Weiping Yan

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Education

University of Minnesota, Twin Cities — **Sep 2026 – Present**
Electrical Engineering and Computer Science, College of Science and Engineering — *Minneapolis, MN*

Delft University of Technology and Eindhoven University of Technology — **Sep 2024 – Aug 2026**
Concurrent study in Computer Science & Engineering, Electrical Engineering, Applied Physics — *Delft & Eindhoven, NL*

- **Honors:** TU/e Honors Academy Honors Bachelor Program; TU Delft Honours Programme Bachelor (HPB) (2025–2026)

Hangzhou Normal University Affiliated High School · Shanghai Foreign Language School — **Sep 2021 – Aug 2024**
High school, science and engineering — *Hangzhou & Shanghai*

Experience

Research Intern, LLM for Electronic Design Automation (EDA) — **Jul 2026 – Sep 2026**
Eastern Institute of Technology, Ningbo — *Ningbo, China*

- Designed and built an independent **LLM-for-EDA** project on circuit understanding and AI-assisted design, from problem formulation and data preparation to LLM-based processing and output evaluation.
- Curated, annotated, cleaned, and quality-checked **circuit-design data** for the group's LLM-for-EDA research, and prepared structured datasets for model development and evaluation.

NLP Research Intern — **Jul 2026 – Sep 2026**
Minnesota NLP Group, University of Minnesota — *Minneapolis, MN*

- Built a pipeline for detecting **computer-use agents** from privacy-preserving browser traces, combining controlled agent runs, reproduction of published detectors, and a tested consent-first **Chrome extension** recorder.
- Wrote the **Python CLI** that validates traces, extracts 23 behavioral features, and runs leave-one-task-out evaluation; **5 PRs** merged into the lab repository.

Co-Founder — **May 2026 – Present**
Veylora Labs — *Minneapolis, MN*

- Building executive-intelligence systems that find patterns across an organization's reports and data and turn them into clear, actionable signals for decision makers.

Research Assistant — **Mar 2026 – Aug 2026**
The Chinese University of Hong Kong — *Shenzhen, China*

- Researched **uncertainty-aware LLMs for generative recommendation** built **Qwen3-8B/LoRA** pipelines for confidence calibration, candidate reranking, and selective recommendation on multi-domain Amazon benchmarks.

Teaching Assistant, Calculus and Logic & Set Theory — **Sep 2025 – Nov 2025**
Eindhoven University of Technology — *Eindhoven, NL*

- Led **tutorial sessions** on limits, derivatives, integrals, formal reasoning, and set operations, and gave structured feedback on students' mathematical reasoning.

Research Assistant — **Jul 2025 – Sep 2025**
Zhangjiang Laboratory — *Shanghai, China*

- Studied and documented the IC design flow and **EDA toolchain**, from schematic design and pre-layout simulation to **DRC/LVS** sign-off and an MPW tape-out workflow overview.

Independent Developer, Computer Science Projects — **May 2024 – Present**
Self-directed study with MIT OpenCourseWare and other open courses — *Remote*

- Implemented **Raft** and sharded key-value services in Go, **xv6** kernel extensions (copy-on-write, mmap), a **TCP/IP** stack in C++, **BusTub** database components (hashing, operators, MVCC), and the **Needle** autodiff and tensor framework (non-credit).

Student Researcher — **Sep 2023 – Feb 2024**
Remote project supervised by Prof. Soummya Kar (Carnegie Mellon University) and Assoc. Prof. Qiang Gao (Southeast University) — *Beijing, China*

- Built a **Python** analysis and visualization pipeline on PM2.5 exposure and respiratory-disease incidence using synthetic data; published in *Theoretical and Natural Science*.

Publications

TempoRAC: Identity-Indexed Local Tempo Routing for Multi-Person Repetitive Action Counting — **Sep 2026**
Weiping Yan*, Jiayu Zhu*, Xu Yan*. *Submitted to ICASSP 2027, under review (equal contribution)*. github.com/appleweiping/TempoRAC

VCH-SegFormer for Street-View Vegetation Mapping and Assessing the Associations Between Urban Vegetation and PM2.5 — **Oct 2026**
Xiaoyan Fei, **Weiping Yan**, Qi Zhu, Xu Yan, Zherong Wu. *Submitted to IEEE Transactions on Geoscience and Remote Sensing, under review*.

RACER-T: Temporal Density Modeling for Pose-Based Repetitive Action Counting — **Oct 2026**
Shuoran Wang, **Weiping Yan**, Xiang Wang. *Submitted to Signal, Image and Video Processing, under review; In Review preprint on Research Square*.

OAM-GA: Reliability-Guided Motion Completion for Occlusion-RobustGaussian Avatars — **2026**
Xiang Wang, Xu Yan, Letian Pei, **Weiping Yan**. *Submitted to DAI 2026, under review*. openreview.net/forum?id=UV2UJ4VHf7

Effect of PM2.5 air pollution on the incidence of respiratory diseases: A Python-based data analysis **Nov 2023**

Weiping Yan. *Theoretical and Natural Science 8(1), 70–75*. 10.54254/2753-8818/8/20240361

Open-Source Contributions

- **20 pull requests merged into 12 projects maintained by others** (Aug–Oct 2026). *Chip design:* OpenRAM, CharLib, HighTide, COmPOSER, ALIGN. *Power systems:* Power Grid Model, power-grid-model-io. *Search and recommendation:* PISA, Pyserini, LensKit. *Speech and NLP:* M*, clulab processors.

Honors and Awards

- **Kaggle Bronze Medal**, Santa 2025 Christmas Tree Packing Challenge: 186th of 3,357 teams (Jan 2026)
- **National First Prize**, Science and Innovation Cup Mathematical Modeling Competition, knowledge track (Dec 2025)
- **First Prize**, National College Student Mathematics Ability Challenge (Dec 2025)
- **First Prize**, CTSS College Student Innovation and Entrepreneurship Competition (Jul 2025)

Certifications

- **Deep Learning Specialization**, DeepLearning.AI (Oct 2026): five courses, from Neural Networks and Deep Learning to Convolutional Neural Networks and Sequence Models
- **Machine Learning Specialization courses**, Stanford Online and DeepLearning.AI (Sep–Oct 2026): Supervised Machine Learning: Regression and Classification · Advanced Learning Algorithms · Unsupervised Learning, Recommenders, Reinforcement Learning

Technical Skills

Languages: Python, C, C++, Java, Rust, TypeScript/JavaScript, MATLAB, Verilog
Tools: Git, NumPy, pandas, Matplotlib, PyTorch, ngspice, LTspice, Origin, Linux
Interests: Artificial intelligence, natural language processing, microelectronics, electronic design automation
Spoken: Mandarin Chinese (native), English (C1)